Type MS · Finding the first term and common difference of an arithmetic series

Answer 1 of 15 · Question 2 part a · 5 marks

2024 NOV · Paper 01 · extract 1 of 10 · find part a in this mark scheme extract

Number		
2 (a)	$9(a+a+d)=a+4d+a+5d+a+6d$ oe $(5a=2d)$ oe $a+2d=12$ oe $5(12-2d)=2d$ oe $d=5 \quad a=2$	M1 B1 M1 A1 A1 [5]
For the main scheme, ALT1 and ALT2, give the first 1 or 2 M marks, regardless of if students carry on to use the method for the 3rd M mark. Favour the ALT which gives the most marks. **ALT3 may not be used except in the circumstances described below – clarify with your team leader if this is not clear.**		
(b)	$\sum_{n=1}^{60} u_n = \frac{60}{2}(2\times"2"+(60-1)\times"5")(=8970)$ $\sum_{n=1}^{14} u_n = \frac{14}{2}(2\times"2"+(14-1)\times"5")(=483)$ or $(u_{14})="2"+(14-1)\times"5"(=67)$ "8970" – "483" = 8487	M1 M1 ddM1 A1 [4]
ALT1	(First term = $u_{15}=$) $"2"+"5"\times(15-1)(=72)$ (Last term = $u_{60}=$) $"2"+"5"\times(60-1)(=297)$ or $"72"+"5"\times(46-1)(=297)$ $\frac{46}{2}("72"+"297")=8487$	M1 M1 ddM1 A1 [4]
ALT2	$(u_{14}=)"2"+"5"\times(14-1)(=67)$ $(u_{60}=)"2"+(60-1)\times"5"(=297)$ $\frac{60}{2}("2"+"297")-\frac{14}{2}("2"+"67")=8487$	M1 M1 ddM1 A1 [4]
This following ALT, may <u>only</u> be used if candidates are clearly and obviously going on to use $\left(\sum_{r=15}^{60} u_r = \right)\frac{46}{2}(2\times"72"+(46-1)\times"5")$ **Allow n = 45 for first 3 method marks.**		
ALT3	(First term =) $"2"+"5"\times(15-1)(=72)$ $\left(\sum_{r=15}^{60} u_r = \right)\frac{46}{2}(2\times"72"+(46-1)\times"5")$	M2 ddM1 A1 [4]
(c)	$2\times\frac{n}{2}(2\times2+(n-1)\times5)-5(5n-3)=10$ oe $(4n+5n^2-5n-25n+15=10)$ $5n^2-26n+5=0$ $(5n-1)(n-5)=0\rightarrow n=\frac{1}{5},5$ $n=5$	M1 A1 M1 A1 [4]
		Total 13 marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 1 of 15 · Question 2 part a · 5 marks

2024 NOV · Paper 01 · extract 2 of 10 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	Correctly forms the equation shown. Any equivalent unsimplified equation. Note: an alternative seen is (the mark is awarded for the second line below or any unsimplified equivalent) $(S_7 - S_4 = 9(a+a+d))$ $\frac{7}{2}(2a+(7-1)) - \frac{4}{2}(2a+(4-1)) = 9(a+a+d)$ $(\Rightarrow 15a = 6d)$
	B1	For the correct equation.
	M1	For a complete and valid method to solve the equations simultaneously, leading to either a = or d =. Allow one processing error **only.** Can be awarded for a complete and valid method to solve any two linear equations, both in a and d, one processing error only. Must lead to either a = or d =
	A1	For either the correct value of a or the correct value of d.
	A1	For the correct values of both a and d.
If candidates use a different method, follow the same principles to award marks or send to Review if this is not clear. Candidates may use any letters or symbols to denote a and d		
(b)	M1	Use of their a and d in the correct formula $\frac{n}{2}(2"a"+(n-1)"d")$ with $n = 60$
	M1	Use of their a and d in the correct formula $\frac{n}{2}(2"a"+(n-1)"d")$ with $n = 14$ Allow n = 15 as a concession.
	ddM1	For their $"\sum_{n=1}^{60} u_n" - "\sum_{n=1}^{14/15} u_n"$. Dependent on both previous method marks. Follow through their value from use of n = 15 if used. The presence of 8970 – 483 will imply M3 here as will 8487.
	A1	For 8487
ALT1	M1	For correct use of $"a"+(n-1)"d"$ with $n = 15$ with their a and their d
	M1	For correct use of $"a"+(n-1)"d"$ with $n = 60$ with their a and their d or correct use of $"a_{15}"+(n-1)"d"$ with $n = 46$ with their d and correctly using their term for a_{15} from the first M mark. Allow n = 45 as a concession.
	ddM1	For correct use of the formula $\frac{n}{2}(u_{15}+u_{60})$. Dependent on both previous method marks. Allow n = 45 as a concession. $\frac{46}{2}("72"+"297")$ or 8487 implies M3
	A1	For 8487
ALT2	M1	For correct use of $"a"+(n-1)"d"$ with $n = 14$ with their a and their d Allow n = 15
	M1	For correct use of $"a"+(n-1)"d"$ with $n = 60$ with their a and their d
	ddM1	For correct use of the formula $\frac{n}{2}(u_1+u_{60}) - \frac{n}{2}(u_{14}+u_{60})$. Dependent on both previous method marks. Allow n = 15 The presence of 8970 – 483 will imply M3 here as will 8487
	A1	For 8487

Type MS · Finding the first term and common difference of an arithmetic series

Answer 1 of 15 · Question 2 part a · 5 marks

2024 NOV · Paper 01 · extract 3 of 10 · find part a in this mark scheme extract

ALT3	M2	For correct use of $"a"+(n-1)"d"$ with $n=15$ with their a and their d
	ddM1	For correct use of the formula $\frac{n}{2}\left(2\times"u_{15}"+(n-1)"d"\right)$. Dependent on both previous method marks. Allow $n = 45$ as a concession.
	A1	8487
(c)	M1	For correct use of $2S_n-5(an+b)=10$ Substitution into S_n must be correct for their values of a and d. **and** candidates must correctly use their a in an expression of the form $an+b$ b can be any integer, $b\neq 0$ $2S_n-5(an+b)=10$ **does not have to be simplified to be awarded this mark.**
	A1	For the correct 3 term quadratic equation = 0. Equivalent coefficients are possible.
	M1	For any valid, complete method to solve their quadratic (see general guidance), which must be of the form $pn^2+qn+r \quad p,q,r\neq 0$ leading to $n=$ $n = 5$ will imply this mark.
	A1	$n = 5$ $n = 0.2$ must be dismissed.

Type MS · Finding the first term and common difference of an arithmetic series

Answer 1 of 15 · Question 2 part a · 5 marks

2024 NOV · Paper 01 · extract 4 of 10 · find part a in this mark scheme extract

Q	Scheme	Marks
3	$\frac{\sin 25}{7}=\frac{\sin B}{10}$ oe $(B=)\sin^{-1}\left(\frac{10\sin 25}{7}\right)$ oe $(B=37.138(254541)$ or $142.861(7455))$	M1dM1
	$(C=180-[(180-37.138(25454))+25])$	
	$(C=)12.138(25451)$	A1 (M1 on ePen)
	$(c^2=)7^2+10^2-2(7)(10)\cos"12.138(25451)"$ or	
	$\frac{c}{\sin"12.138(25451)"}=\frac{7}{\sin 25}$ oe eg $(c=)\frac{7}{\sin 25}\times\sin"12.138(25451)"$ or	
	$\frac{c}{\sin"12.138(25451)"}=\frac{10}{\sin"142.861(7455)"}$ oe	
	eg $(c=)\frac{10}{\sin"142.861(7455)"}\times\sin"12.138(25451)"$	
	**allow students to follow through using their acute angle for B	ddM1
	$(c^2=)7^2+10^2-2(7)(10)\cos"117.861(7455)"$ or	
	$\frac{c}{\sin"117.861(7455)"}=\frac{7}{\sin 25}$ oe eg $(c=)\frac{7}{\sin 25}\times\sin"117.861(7455)"$ or	
	$\frac{c}{\sin"117.861(7455)"}=\frac{10}{\sin"37.138(254541)"}$ oe	
	eg $(c=)\frac{10}{\sin"37.138(254541)"}\times\sin"117.861(7455)"$	
	$(c=)3.5(\text{cm})$	A1 [5]
ALT1	$7^2=10^2+(AB)^2-2(10)(AB)\cos 25$	M1
	$0=(AB)^2-20\cos 25(AB)+100-49$ $(20\cos 25=18.12615574)$	dM1
	$0=(AB)^2-20\cos 25(AB)+51$	A1 (M1 on ePen)
	$(AB=)\frac{"--20\cos 25"\pm\sqrt{("-20\cos 25")^2-4("1")("51")}}{2\times"1"}(=3.48, 14.64)$	ddM1
	$(c=)3.5(\text{cm})$	A1 [5]
ALT2	M1 M1 as main scheme (then students are working in right angled triangles).	
	$(AX=)10\cos 25$ or $9.(06307787)$	B1 (M1 on ePen)
	$(BX=)7\sin("37.14.....")\Rightarrow 10\cos 25-7\sin("37.14.....")$	ddM1
	or if working with acute angle	
	$(BX=)7\sin("37.14.....")\Rightarrow 10\cos 25+7\sin("37.14.....")$	A1 [5]
	$=3.5((cm))$	
		Total 5 marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 1 of 15 · Question 2 part a · 5 marks

2024 NOV · Paper 01 · extract 5 of 10 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
	M1	For a fully correct substitution into the sine rule.
	dM1	For a correct rearrangement to find angle B, implied by sight of the correct acute or obtuse angle. Dependent on previous method mark.
	A1 (M1 on ePen)	For angle C, accept angles which round to 12.
	ddM1	Correct substitution into the cosine rule or sine rule using their angles B and/or C. Allow angle B to be an acute angle and angle C to be the angle that comes from this acute angle using the sum of angles in a triangle. This mark is dependent on the first 2 method marks and is for correct use of the cosine or sine rule, given they have found an angle for B and used this to find angle C.
	A1	For awrt 3.5 (cm)
ALT1	M1	For a fully correct substitution into the cosine rule. Allow AB to be denoted by any letter
	dM1	For a fully correct rearrangement = 0, dependent on the previous method mark.
	A1 (M1 on ePen)	For the correct 3 term quadratic equation = 0.
	ddM1	For a minimally acceptable attempt to solve their quadratic – see general guidance. Dependent on both previous method marks. Sight of 3.48….. or 14.64 may imply this mark
	A1	For awrt 3.5 (cm) Must reject any second solution.
ALT2	M2	As first 2 marks in main scheme
	B1 (M1 on ePen)	For AX either 10cos25 or answers which round to 9
	ddM1	For finding BX correctly and a correct method to find AB, allowing use of acute B
	A1	For awrt 3.5 (cm)

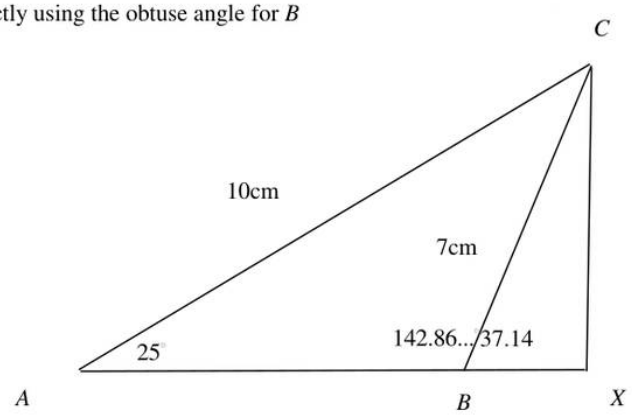
Type MS · Finding the first term and common difference of an arithmetic series

Answer 1 of 15 · Question 2 part a · 5 marks

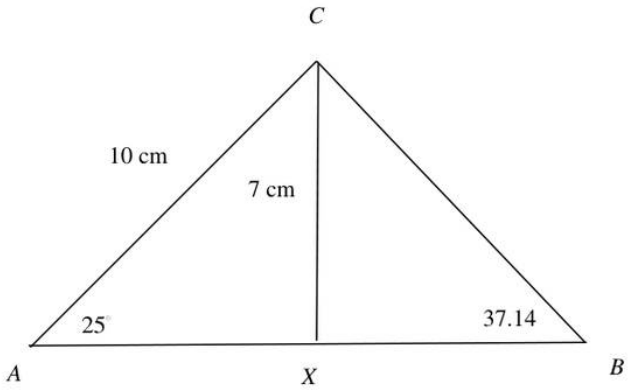
2024 NOV · Paper 01 · extract 6 of 10 · find part a in this mark scheme extract

Particularly to help with ALT2:

If correctly using the obtuse angle for B



Using the acute angle for B



Type MS · Finding the first term and common difference of an arithmetic series

Answer 1 of 15 · Question 2 part a · 5 marks

2024 NOV · Paper 01 · extract 7 of 10 · find part a in this mark scheme extract

Question number	Scheme	Marks
	Remember to look for work on the diagram.	
4	$(FD)=\sqrt{15^2+20^2}$ $(FD=)25$ $(BF=)15\sin 40$ oe $(\Rightarrow (BF)=9.641814145)$ $\sin BDF=\frac{"15\sin 40"}{"\sqrt{15^2+20^2}"}(=0.3856725658)$ or $\sin^{-1}\left(\frac{"9.641814145"}{"25"}\right)$ $(BDF)=22.7°$	M1 A1 M1 M1 dM1 A1 [6]
Example of an ALT	$(DC=)15\times\cos 40$ and $(BD=)\sqrt{20^2+("15\cos 40")^2}$ $(BD=)$awrt 23.1 $(DC=)15\times\cos 40$ and $(FB=)\sqrt{15^2-(15\cos 40)^2}$ oe $(\Rightarrow (FB)=9.641814145)$ $\tan BDF=\frac{"\sqrt{15^2-(15\cos 40)^2}"}{"\sqrt{20^2+("15\cos 40")^2}"}(=0.4180117433)$ or $\tan^{-1}\left(\frac{"9.641814145"}{"23.1"}\right)$ $(BDF)=22.7°$	M1 A1 M1 dM1 M1 A1 [6]
		Total 6 marks

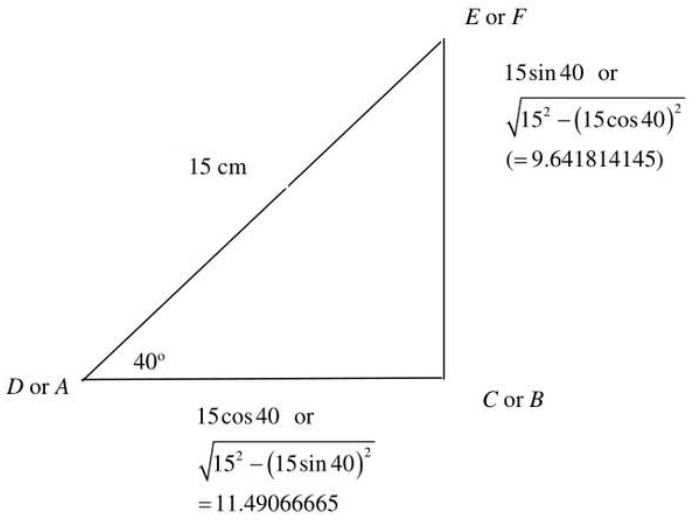
Part	Mark	Additional Guidance
$(BF=)15\sin 40$ oe $(\Rightarrow (BF)=9.641814145)$ means does not need to be labelled *BF* but there mustn't be anything to contradict this. (General principle of marking) It is fine to label with equivalent sides eg (*EC* = *FB*) There are a number of ways candidates are completing this question. Please mark to the following general principles – send anything to review that can't be marked using these.		
	M1	For a **full** and correct method to find *FD* or *BD*
	A1	For *FD* = 25 or *BD* = awrt 23.1
	M1	For a **full** and correct method to find a second side in triangle *BDF*
	M1	For correctly identifying the correct angle to be found. Dependent on previous method mark. Look for any clear identification that this is the angle they're finding, **including on the diagram** or in written work.
	dM1	For the use of their values for two sides in triangle *BDF* in a correct trig **equation** or the correct **expression** to find the angle with the relevant inverse trig function. There must be a full, complete and correct **method** which leads to an equation or expression to find angle *BDF*, part methods should not be awarded this mark. Dependent on the previous method mark only ie the correct angle identified, which may be implied in a correct equation or expression. If it is clear candidates are using 2 of their values for *BF*, *FD* and *BD*, this mark can be awarded, even if the work to calculate either is incorrect. $\sin BDF=\left(\frac{"BF"}{"DF"}\right)$ or $\cos BDF=\left(\frac{"BD"}{"DF"}\right)$ or $\tan BDF=\left(\frac{"BF"}{"BD"}\right)$ or $\sin^{-1}\left(\frac{"BF"}{"DF"}\right)$ or $\cos^{-1}\left(\frac{"BD"}{"DF"}\right)$ or $\tan^{-1}\left(\frac{"BF"}{"BD"}\right)$ If incorrect lengths are found, you may allow $\left(\frac{"BF"}{"DF"}\right)$ or $\left(\frac{"BD"}{"DF"}\right)>1$ for this mark Note: if students choose to use cosine rule, they must additionally use a full and correct method to find a third side of triangle *BDF* and have either of these $\cos BDF=\frac{DF^2+BD^2-BF^2}{2\times DF\times BD}$ or $\cos^{-1}\left(\frac{DF^2+BD^2-BF^2}{2\times DF\times BD}\right)$
	A1	Any answer which rounds to (awrt) 22.7°

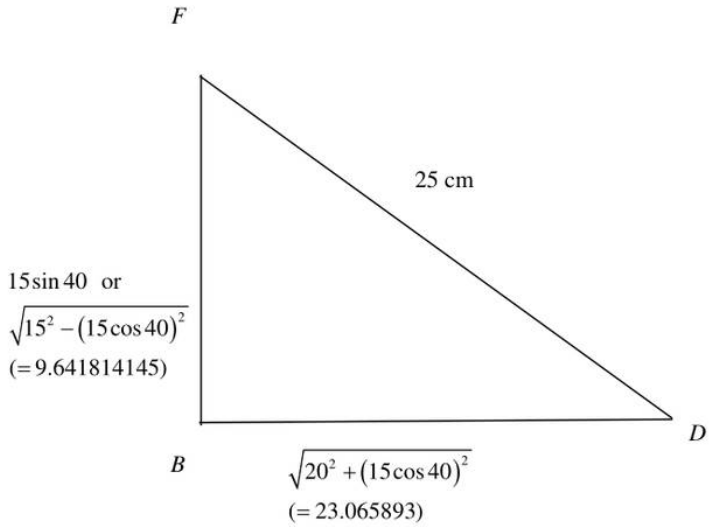
Type MS · Finding the first term and common difference of an arithmetic series

Answer 1 of 15 · Question 2 part a · 5 marks

2024 NOV · Paper 01 · extract 8 of 10 · find part a in this mark scheme extract

Useful triangles and sides.





Type MS · Finding the first term and common difference of an arithmetic series

Answer 1 of 15 · Question 2 part a · 5 marks

2024 NOV · Paper 01 · extract 9 of 10 · find part a in this mark scheme extract

Question number	Scheme	Marks
5 (a)	$3t^2-16t+5=0 \rightarrow (3t-1)(t-5) \rightarrow t =$ $t=\frac{1}{3}$ oe $\quad t=5$	M1 M1 A1 [3]
(b)	$(a=)6t-16>0$ $t>\frac{8}{3}$ oe	M1 A1 [2]
(c)	$\int_{(1)}^{(4)}(3t^2-16t+5)\,dt$ $\left[t^3-8t^2+5t\right]_{(1)}^{(4)}$ $\left(4^3-8(4)^2+5(4)\right)-\left(1^3-8(1)^2+5(1)\right)$ -42 (distance) = 42	 M1 A1 M1 A1 A1 [5]
		Total 10 marks

Part	Mark	Additional Guidance
(a)	M1	For setting $v = 0$ and attempting to solve, see general guidance for definition of a minimum attempt to solve leading to $t =$
	M1	For a correct value of t. Any correct value will imply M2.
	A1	For both correct values of t Both correct values will imply M2 A1
(b)	M1	For an attempt to differentiate and setting their expression > 0 (allow placing = 0 as a concession) See general guidance for definition of a valid attempt. Additionally for this mark, no power of t to increase, at least one term differentiated correctly.
	A1	For $t>\frac{8}{3}$ oe Accept rounding to 1 dp or better eg 2.7, 2.67, 2.667 etc Or indication of recurring e.g. $2.\dot{6}$ or $2.6'$ or 2.6... minimum 3 dots
(c)	M1	For an attempt to integrate the given expression for v, see general guidance for the definition of a valid attempt. Additionally, no power of t to decrease. Limits do not need to be present.
	A1	For a correct integration. Limits do not need to be present.
	M1	For correct substitution of limits into **their** changed expression. Can be any changed expression with a minimum of 2 terms. This mark may be implied by sight of – 42 or 42 or – 44 – – 2 or 44 – 2. If the final answer is incorrect, must see substitution of both limits correctly at least once. For this mark, students may assume an arbitrary constant (typically $C = 0$) or leave an arbitrary constant in their working and have the equivalent $\left(4^3-8(4)^2+5(4)+C\right)-\left(1^3-8(1)^2+5(1)+C\right)$ Candidates cannot attain this mark for just – 44 and – 2 without a subtraction.
	A1	For – 42 or 42.
	A1	Correct positive distance given.
Any candidate not showing the step of integration – 0 marks for this part of the question.		

Type MS · Finding the first term and common difference of an arithmetic series

Answer 1 of 15 · Question 2 part a · 5 marks

2024 NOV · Paper 01 · extract 10 of 10 · find part a in this mark scheme extract

Question number	Scheme	Marks
6(a)	eg $\frac{a}{\sqrt{4\left(1+\frac{bx}{4}\right)}}$ or $\frac{a}{\sqrt{4}\sqrt{\left(1+\frac{bx}{4}\right)}}$ or $\frac{a}{4^{\frac{1}{2}}\left(1+\frac{bx}{4}\right)^{\frac{1}{2}}}$ or $a\left(4^{-\frac{1}{2}}\right)\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$ or $a\left(4\left(1+\frac{bx}{4}\right)\right)^{-\frac{1}{2}}$ $=\frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$	M1 A1*cso [2]
(b)	$\left(\frac{a}{2}\right)\left[1+\left(-\frac{1}{2}\right)\left(\frac{bx}{4}\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)}{2!}\left(\frac{bx}{4}\right)^2+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\left(-\frac{1}{2}-2\right)}{3!}\left(\frac{bx}{4}\right)^3\right]$ oe $\left(\left(\frac{a}{2}\right)\left[1+\left(-\frac{bx}{8}\right)+\frac{3}{128}b^2x^2+\left(-\frac{5}{1024}b^3x^3\right)\right]\right)$ Not necessary - but must be checked if seen All intermediate steps must be checked for errors $\frac{a}{2}-\frac{ab}{16}x+\frac{3ab^2}{256}x^2-\frac{5ab^3}{2048}x^3$ $P=\frac{a}{2}, Q=-\frac{ab}{16}*, R=\frac{3ab^2}{256}, S=-\frac{5ab^3}{2048}*$	M1 M1 A1 A1*cso [4]
(c)	$-\frac{ab}{16}=\frac{128}{5}.\left(-\frac{5ab^3}{2048}\right)$ or $\frac{5}{128}.\left(-\frac{ab}{16}\right)=-\frac{5ab^3}{2048}$ or $-\frac{5ab}{2048}=-\frac{5ab^3}{2048}$ or $-\frac{ab}{16}=-\frac{ab^3}{16}$ $\left(b^2=1\right)b=1*$	M1 A1*

Type MS · Finding the first term and common difference of an arithmetic series

Answer 2 of 15 · Question 6 part b · 2 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

6(a)	$a = S_1 = 1(15+2\times1) = 17$	B1	
	$S_2 = 2(15+2\times2)(=38) = 2a+d$	M1A1	
	$2\times17+d=38 \Rightarrow d=4$	A1	(4)
(b)	20th term $= a+19d = 17+19\times4 = 93$	M1A1	(2)
(c)	$S_{2p}-2S_p = 1+S_{p-1}$		
	$2p(15+4p)-2p(15+2p) = 1+(p-1)(13+2p)$	M1	
	$2p^2-11p+12=0$	A1	
	$(2p-3)(p-4)=0 \Rightarrow p=4 \left(p\neq\frac{3}{2};\ \text{may not be seen}\right)$	M1A1	(4)
			[10]

Part	Mark	Notes
(a)	B1	For the first term $a = 17$ $\left[a = S_1 = 1(15+2\times1) = 17\right]$
	M1	For the second term. Uses the given summation formula to form a linear equation in a and d for a minimally acceptable response of $k = 2a+d$ where k is a positive integer.
	A1	For the correct linear equation $38 = 2a+d$
	A1	For the correct value of $d = 4$
	ALT 1	
	B1	For the first term $a = 17$
	M1	For using a correct summation formula $n(15+2n) = \frac{n}{2}\left(2a+\left[n-1\right]d\right) \Rightarrow 30+2n = 2a-d+nd$ and equates coefficients
	A1	For equating coefficients of n $4n = dn \Rightarrow d = \ldots$ and $30 = 2a-4 \Rightarrow a = \ldots$
		For the correct value of $d = 4$

Type MS · Finding the first term and common difference of an arithmetic series

Answer 2 of 15 · Question 6 part b · 2 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

	ALT 2	
	B1	For the first term $a = 17$
	M1	Uses two values of n to set up a pair of simultaneous equations.e.g. $S_4 = 4(15+2\times4) = 92$ and $92 = \frac{4}{2}(2a+3d) \Rightarrow 46 = 2a+3d$ $S_5 = 5(15+2\times5) = 125$ and $125 = \frac{5}{2}(2a+4d) \Rightarrow 50 = 2a+4d$
	A1	Attempts to solve the pair of equations
	A1	$d = 4$
(b)	M1	For using the correct nth term formula with **their a and their d** $U_{20} = '17'+19\times'4' = ...$
	A1	For the correct 20th term = 93
(c)	M1	Uses the given summation formula with the correct substitution $2p(15+4p)-2p(15+2p) = 1+(p-1)(13+2p)$
	A1	For achieving the correct 3TQ $2p^2-11p+12 = 0$
	ALT	
	M1	Uses the summation formula: **Follow through their a and d** $S_{2p} = \frac{2p}{2}(2\times17+(2p-1)4) = p(30+8p)$ $2S_p = 2\times\frac{p}{2}(2\times17+(p-1)4) = p(30+4p)$ $S_{p-1} = \frac{p-1}{2}(2\times17+(2[p-1]-1)4) = (p-1)(13+2p)$ For a correct substitution into the given expression $p(30+8p)-p(30+4p) = 1+(p-1)(13+2p)$ oe
	A1	For achieving the correct 3TQ $2p^2-11p+12 = 0$
	M1	For attempting to solve their 3TQ (provided it is a 3TQ) by any valid method. $2p^2-11p+12 = (2p-3)(p-4) = 0 \Rightarrow p = ..., ...$
	A1	For $p = 4$ If they give both roots of their 3TQ as an answer without rejecting $p = 1.5$ A0

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Type MS · Finding the first term and common difference of an arithmetic series

Answer 2 of 15 · Question 6 part b · 2 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question Number	Scheme	Marks
	$\left(\quad 9\right)^2 \quad 81 \quad \left(\quad 9\right)^2 \quad 25$	

Type MS · Finding the first term and common difference of an arithmetic series

Answer 3 of 15 · Question 1 part whole · 4 marks

2023 JAN · Paper 01 · find part whole in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Finding the first term and common difference of an arithmetic series

Answer 4 of 15 · Question 7 part a.i · 0 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

Number		
7 (a)(i)	$a+4d=4x+6$ $a+7d=7x+3$ $\Rightarrow 3d=3x-3\Rightarrow d=x-1$*	M1A1cso
(ii)	$a+7(x-1)=7x+3\Rightarrow a=10$ or $a+4(x-1)=4x+6\Rightarrow a=10$	M1A1 [4]
(b)	$42=10+8(x-1)\Rightarrow x=5$	M1A1 [2]
(c)	$d=x-1\Rightarrow d=5-1=4$ $S_{n+1}=12U_n+18\Rightarrow \frac{n+1}{2}\left(2\times 10+\left[(n+1)-1\right]4\right)=12\left[10+(n-1)4\right]+18$ $\Rightarrow n^2-18n-40=0$ $\Rightarrow (n-20)(n+2)=0\Rightarrow n=20$	B1 M1 M1 M1A1 [5]
		Total 11 marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 4 of 15 · Question 7 part a.i · 0 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 5 of 15 · Question 7 part a.ii · 4 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

Number		
7 (a)(i)	$a+4d=4x+6$ $a+7d=7x+3$ $\Rightarrow 3d=3x-3\Rightarrow d=x-1$*	M1A1cso
(ii)	$a+7(x-1)=7x+3\Rightarrow a=10$ or $a+4(x-1)=4x+6\Rightarrow a=10$	M1A1 [4]
(b)	$42=10+8(x-1)\Rightarrow x=5$	M1A1 [2]
(c)	$d=x-1\Rightarrow d=5-1=4$ $S_{n+1}=12U_n+18\Rightarrow \frac{n+1}{2}\left(2\times 10+\left[(n+1)-1\right]4\right)=12\left[10+(n-1)4\right]+18$ $\Rightarrow n^2-18n-40=0$ $\Rightarrow (n-20)(n+2)=0\Rightarrow n=20$	B1 M1 M1 M1A1 [5]
		Total 11 marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 5 of 15 · Question 7 part a.ii · 4 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 6 of 15 · Question 7 part b · 2 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Number		
7 (a)(i)	$a+4d=4x+6$ $a+7d=7x+3$ $\Rightarrow 3d=3x-3\Rightarrow d=x-1$*	M1A1cso
(ii)	$a+7(x-1)=7x+3\Rightarrow a=10$ or $a+4(x-1)=4x+6\Rightarrow a=10$	M1A1 [4]
(b)	$42=10+8(x-1)\Rightarrow x=5$	M1A1 [2]
(c)	$d=x-1\Rightarrow d=5-1=4$	B1
	$S_{n+1}=12U_n+18\Rightarrow \frac{n+1}{2}\left(2\times 10+\left[(n+1)-1\right]4\right)=12\left[10+(n-1)4\right]+18$	M1
	$\Rightarrow n^2-18n-40=0$	M1
	$\Rightarrow (n-20)(n+2)=0\Rightarrow n=20$	M1A1 [5]
		Total 11 marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 6 of 15 · Question 7 part b · 2 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 7 of 15 · Question 5 part b · 2 marks

2024 JUN · Paper 01 (regional R) · find part b in this mark scheme extract

5(a)	**Mark both parts together** $4k+1=a+(6-1)d \Rightarrow [\ 4k+1=a+(6-1)d\ ]$ $36k+1=\frac{10}{2}(2a+9d) \Rightarrow [36k+1=10a+45d]$ Solves the simultaneous equations by any method: **(i)** $d=\frac{4k+9}{5}$ **(ii)** $a=-8$*	M1 M1 M1 A1 A1 cso [5]
(b)	$7=-8+3d \Rightarrow 7=-8+3\left(\frac{4k+9}{5}\right) \Rightarrow 25=4k+9 \Rightarrow k=4$ *	M1A1 cso [2]
(c)	$\left[d=\frac{4\times4+9}{5}=5\right.$ you may see this in part (c)$\left.\right]$ $\frac{n}{2}(2\times-8+(n-1)\times5)=5[-8+(n+10-1)5]+105$ $\Rightarrow -21n+5n^2=370+50n+210 \Rightarrow 5n^2-71n-580=0$ $\Rightarrow (5n+29)(n-20)=0 \Rightarrow n=20$	M1 dM1 M1A1 [4]
		Total 11 marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 8 of 15 · Question 1 part a.i · 0 marks

2020 JAN · Paper 01 · extract 1 of 3 · find part a.i in this mark scheme extract

1 (a) (i)	$a+d+a+8d=0$ $\quad a+3d+a+5d+a+9d=14$ Solve simultaneously $d=4$	M1 M1 A1
(ii)	$a=-18$	A1 (4)
(b)	$\frac{3n}{2}[48+6(n-1)]=\frac{2n}{2}[48+6(2n-1)]$ $3n(42+6n)=2n(42+12n) \Rightarrow 6n^2-42n=0$ $6n(n-7)=0 \Rightarrow n=[0,\ 7]$ $n=7$	M1 A1 A1 M1 A1 (5) **[9]**

Type MS · Finding the first term and common difference of an arithmetic series

Answer 8 of 15 · Question 1 part a.i · 0 marks

2020 JAN · Paper 01 · extract 2 of 3 · find part a.i in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For writing down both correct expressions in terms of a and d $a+d+a+8d=0$ $\quad$ $a+3d+a+5d+a+9d=14$
	M1	For attempting to solve their simultaneous equations for a and d $2a+9d=0$ $3a+17d=14$
	A1 (i)	For $d=4$ * **This is a show question – there must be no errors for the award of this mark**
	B1 (ii)	For $a=-18$ **This is an A mark in Epen**
(b)	M1	For the **correct** use of the **correct** summation formula on **one of** the LHS or the RHS of the following equation. $\frac{3n}{2}\left(2\times24+6[n-1]\right)=\frac{2n}{2}\left(2\times24+6[2n-1]\right)$ No simplification is required for this mark.
	A1	For a fully correct equation as shown above – simplified or unsimplified
	A1	For reaching a correct 2TQ equation in n $126n+18n^2=84n+24n^2 \Rightarrow 6n^2-42n=0$
	M1	For attempting to solve **their** quadratic (See General Guidance for the definition of an attempt) $6n^2-42n=0 \Rightarrow 6n(n-7)=0 \Rightarrow n=[0,\ 7]$
	A1	$n=7$ Condone the value of 0 for this mark
	ALT	
	M1	For the **correct** use of the **correct** summation formula on **one of** the LHS or the RHS of the following equation. $\frac{3n}{2}\left(2\times24+6[n-1]\right)=\frac{2n}{2}\left(2\times24+6[2n-1]\right)$ No simplification is required for this mark.
	A1	For a fully correct equation as shown above – simplified or unsimplified
	A1	Divides through n to reach a linear equation to give $6n=42$ oe
	M1	Solves their linear equation in n
	A1	$n=7$

Type MS · Finding the first term and common difference of an arithmetic series

Answer 8 of 15 · Question 1 part a.i · 0 marks

2020 JAN · Paper 01 · extract 3 of 3 · find part a.i in this mark scheme extract

Paper 1		
Question number	Scheme	Marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 9 of 15 · Question 1 part a.ii · 4 marks

2020 JAN · Paper 01 · extract 1 of 3 · find part a.ii in this mark scheme extract

1 (a) (i)	$a+d+a+8d=0 \qquad a+3d+a+5d+a+9d=14$ Solve simultaneously $d=4$	M1 M1 A1
(ii)	$a=-18$	A1 (4)
(b)	$\frac{3n}{2}[48+6(n-1)]=\frac{2n}{2}[48+6(2n-1)]$ $3n(42+6n)=2n(42+12n) \Rightarrow 6n^2-42n=0$ $6n(n-7)=0 \Rightarrow n=[0,\ 7]$ $n=7$	M1 A1 A1 M1 A1 (5) **[9]**

Type MS · Finding the first term and common difference of an arithmetic series

Answer 9 of 15 · Question 1 part a.ii · 4 marks

2020 JAN · Paper 01 · extract 2 of 3 · find part a.ii in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For writing down both correct expressions in terms of a and d $a+d+a+8d=0$ $\quad$ $a+3d+a+5d+a+9d=14$
	M1	For attempting to solve their simultaneous equations for a and d $2a+9d=0$ $3a+17d=14$
	A1 (i)	For $d=4$ * **This is a show question – there must be no errors for the award of this mark**
	B1 (ii)	For $a=-18$ **This is an A mark in Epen**
(b)	M1	For the **correct** use of the **correct** summation formula on **one of** the LHS or the RHS of the following equation. $\frac{3n}{2}\left(2\times24+6\left[n-1\right]\right)=\frac{2n}{2}\left(2\times24+6\left[2n-1\right]\right)$ No simplification is required for this mark.
	A1	For a fully correct equation as shown above – simplified or unsimplified
	A1	For reaching a correct 2TQ equation in n $126n+18n^2=84n+24n^2\Rightarrow 6n^2-42n=0$
	M1	For attempting to solve **their** quadratic (See General Guidance for the definition of an attempt) $6n^2-42n=0\Rightarrow 6n\left(n-7\right)=0\Rightarrow n=\left[0,\ 7\right]$
	A1	$n=7$ Condone the value of 0 for this mark
	ALT	
	M1	For the **correct** use of the **correct** summation formula on **one of** the LHS or the RHS of the following equation. $\frac{3n}{2}\left(2\times24+6\left[n-1\right]\right)=\frac{2n}{2}\left(2\times24+6\left[2n-1\right]\right)$ No simplification is required for this mark.
	A1	For a fully correct equation as shown above – simplified or unsimplified
	A1	Divides through n to reach a linear equation to give $6n=42$ oe
	M1	Solves their linear equation in n
	A1	$n=7$

Type MS · Finding the first term and common difference of an arithmetic series

Answer 9 of 15 · Question 1 part a.ii · 4 marks

2020 JAN · Paper 01 · extract 3 of 3 · find part a.ii in this mark scheme extract

Paper 1		
Question number	Scheme	Marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 10 of 15 · Question 2 part a.i · 0 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 3 · find part a.i in this mark scheme extract

2 (a)(i)	$a+4d=46 \qquad a+19d=181$ oe	B1
	$"15d=135" \Rightarrow d=9$	M1 A1 [3]
(a)(ii)	$a=10$	B1 [1]
(b)	$\sum_{n=1}^{50} u_n = 25(2\times"10"+(50-1)\times"9") = 11525$	M1
	$\sum_{n=1}^{20} u_n = 10(2\times"10"+(20-1)\times"9") = 1910$	M1
	$"11525"-"1910" = 9615$	ddM1 A1 [4]
ALT	(First term =) $"10"+"9"\times(21-1) = 190$	{M1}
	(Last term =) $"10"+"9"\times(50-1) = 451$	{M1}
	$\frac{30}{2}("190"+"451") = 9615$	{ddM1}{A1} [4]
		Total 8 marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 10 of 15 · Question 2 part a.i · 0 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 3 · find part a.i in this mark scheme extract

Type MS · Finding the first term and common difference of an arithmetic series

Answer 10 of 15 · Question 2 part a.i · 0 marks

2022 JAN · Paper 01 (regional R) · extract 3 of 3 · find part a.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 11 of 15 · Question 2 part a.ii · 4 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 3 · find part a.ii in this mark scheme extract

2 (a)(i)	$a+4d=46 \qquad a+19d=181$ oe	B1
	$"15d=135" \Rightarrow d=9$	M1 A1 [3]
(a)(ii)	$a=10$	B1 [1]
(b)	$\sum_{n=1}^{50} u_n = 25(2\times"10"+(50-1)\times"9") = 11525$	M1
	$\sum_{n=1}^{20} u_n = 10(2\times"10"+(20-1)\times"9") = 1910$	M1
	$"11525"-"1910" = 9615$	ddM1 A1 [4]
ALT	(First term =) $"10"+"9"\times(21-1) = 190$	{M1}
	(Last term =) $"10"+"9"\times(50-1) = 451$	{M1}
	$\frac{30}{2}("190"+"451") = 9615$	{ddM1}{A1} [4]
		Total 8 marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 11 of 15 · Question 2 part a.ii · 4 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 3 · find part a.ii in this mark scheme extract

Type MS · Finding the first term and common difference of an arithmetic series

Answer 11 of 15 · Question 2 part a.ii · 4 marks

2022 JAN · Paper 01 (regional R) · extract 3 of 3 · find part a.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 12 of 15 · Question 8 part a · 6 marks

2023 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks
8(a)	$42 = \frac{4}{2}\left(2a + (4-1)d\right) \Rightarrow 42 = 4a + 6d$ oe	B1
	$23 = a + 4d$	B1
	$\Rightarrow a = 3, \quad d = 5$	M1A1
	$n\text{th term} = 3 + (n-1)5 = 5n - 2 \Rightarrow S_n = \sum_{r=1}^{n}(5r-2)$	M1A1
	$[P = 5, \ Q = 2]$	[6]
(b)	$S_{2n} - 3U_n = 1062$	
	$\Rightarrow \frac{2n}{2}\left(2\times'3' + (2n-1)'5'\right) - 3\left['3' + (n-1)'5'\right] = 1062$	M1
	$\Rightarrow 10n^2 + n - 15n + 6 = 1062 \Rightarrow 10n^2 - 14n - 1056 = 0$	M1
	$10n^2 - 14n - 1056 = (5n + 48)(2n - 22) = 0$	M1
	$\Rightarrow n = 11$	A1
		[4]
		Total 10 marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 12 of 15 · Question 8 part a · 6 marks

2023 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	Forms a correct equation in a and d for **either** the sum of the first 4 terms or for the 5th term This must be correct.
	B1	Forms correct equations in a and d for **both** the sum of the first 4 terms and for the 5th term Both must be correct.
	M1	Solves their two equations simultaneously by any method. Accept a pair of equations as follows: $pa+qd=21$ or 42 $ra+sd=23$ where $p\neq r$ and $q\neq s$ Allow a maximum of one arithmetical error in the solution of their SE.
	A1	For **both** $a=3$ and $d=5$
	M1	Attempts to form the required expression using their values of a and d They must use the nth term for this.
	A1	For the correct expression as written. This must be exactly as written in the question with the inclusion of the $\sum$. However, allow the omission of $S_n=.....$
	ALT for last 2 marks	
	M1	$S_n=\sum_{r=1}^{n}(Pr-Q)\Rightarrow 3=P-Q \quad 5=(2P-Q)-(P-Q)=P$ $\Rightarrow 3=5-Q\Rightarrow Q=2$
	A1	For the correct expression as written. This must be exactly as written in the question with the inclusion of the $\sum$. However, allow the omission of $S_n=.....$ Allow also $S_n=\sum_{r=1}^{n}(Pr-Q) \quad (\Rightarrow) \quad P=5, Q=2$ for this mark.
(b)	M1	Applies the summation formula and the nth term formula correctly to form an equation in n using their values for a and d For this mark you must see an equation, e.g., $\frac{2n}{2}(2\times 3+(2n-1)5)-3[3+(n-1)5]=1062$ However, allow recovery in the next step.
	M1	Forms a 3TQ $\quad 10n^2-14n-1056=0$ o.e. Condone missing = 0 if it is clear they are solving this equation.
	M1	Attempts to solve their 3TQ Accept any method. If their 3TQ is incorrect, only award this mark when you can see their method. If a calculator is used with an incorrect 3TQ and no method is seen, this mark is not available.

Type MS · Finding the first term and common difference of an arithmetic series

Answer 12 of 15 · Question 8 part a · 6 marks

2023 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

		Note: We do not need to see $-\frac{48}{5}$. Some candidates will automatically reject this solution.
	A1	For the correct value of n Do not award if $-\frac{48}{5}$ is included as a value.

Question	Scheme	Marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 13 of 15 · Question 1 part whole · 5 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 3 · find part whole in this mark scheme extract

1	**Method 1** $16 = a + (5-1)d \quad (= a + 4d)$ or $301 = a + (100-1)d \quad (= a + 99d)$	M1
	$95d = 285$	M1
	$d = 3 \qquad a = 4$	A1
	Method 2 $301 - 16 = 295 \Rightarrow 295 = (100-5)d$	[M1
	$\Rightarrow d = \frac{285}{95} = (3)$	M1
	$d = 3 \quad a = 4$	A1]
	Sum to 50 terms Uses $S_n = \frac{n}{2}(2a + (n-1)d)$	
	$S_{50} = \frac{50}{2}[2\times"4"+"3"(50-1)] = 3875$ **ALT**	M1A1
	Uses $S_n = \frac{n}{2}(a+l)$	[5]
	50th term = "4" + 49 ×"3" (=151) **and** $S_{50} = \frac{50}{2}('4'+'151') = 3875$	[M1A1]
		Total 5 marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 13 of 15 · Question 1 part whole · 5 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 3 · find part whole in this mark scheme extract

Mark	Additional Guidance
Method 1	
M1	At least, one fully correct equation.
M1	Solves their linear equations simultaneously, allow one arithmetical or processing slip to eliminate a (or possibly d).
A1	Both a and d correct.
Method 2	
M1	Uses the difference of the two terms $(301-16)$ together with the difference in position $(100-5)$ and equates to form the equation $285=95d$ This may be implied by their working.
M1	Solves the equation, allow one arithmetical or processing slip to find a value for d
A1	Both a and d correct.
NOTE: If you see the correct values of a and d that come from a method without errors, award **M1M1A1**	
Summation	
M1	Substitution of their values for a and d into the correct summation formula. **OR** They must use the formula for the 50th term correctly using their values and use the correct formula for first plus last with their values.
A1	For 3875

Type MS · Finding the first term and common difference of an arithmetic series

Answer 13 of 15 · Question 1 part whole · 5 marks

2022 JUN · Paper 01 (regional R) · extract 3 of 3 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 14 of 15 · Question 5 part a.i · 0 marks

2024 JUN · Paper 01 (regional R) · find part a.i in this mark scheme extract

5(a)	**Mark both parts together**	
	$4k+1=a+(6-1)d \Rightarrow [\ 4k+1=a+(6-1)d\ ]$	M1
	$36k+1=\frac{10}{2}(2a+9d) \Rightarrow [36k+1=10a+45d]$	M1
	Solves the simultaneous equations by any method:	M1
(i)	$d=\frac{4k+9}{5}$	A1
(ii)	$a=-8$*	A1 cso [5]
(b)	$7=-8+3d \Rightarrow 7=-8+3\left(\frac{4k+9}{5}\right) \Rightarrow 25=4k+9 \Rightarrow k=4$ *	M1A1 cso [2]
(c)	$\left[d=\frac{4\times4+9}{5}=5 \quad \text{you may see this in part (c)}\right]$	
	$\frac{n}{2}(2\times-8+(n-1)\times5)=5[-8+(n+10-1)5]+105$	M1
	$\Rightarrow -21n+5n^2=370+50n+210 \Rightarrow 5n^2-71n-580=0$	dM1
	$\Rightarrow (5n+29)(n-20)=0 \Rightarrow n=20$	M1A1 [4]
		Total 11 marks

Type MS · Finding the first term and common difference of an arithmetic series

Answer 15 of 15 · Question 5 part a.ii · 5 marks

2024 JUN · Paper 01 (regional R) · find part a.ii in this mark scheme extract

5(a)	**Mark both parts together** $4k+1=a+(6-1)d \Rightarrow [\ 4k+1=a+(6-1)d\ ]$ $36k+1=\frac{10}{2}(2a+9d) \Rightarrow [36k+1=10a+45d]$ Solves the simultaneous equations by any method: **(i)** $d=\frac{4k+9}{5}$ **(ii)** $a=-8$*	M1 M1 M1 A1 A1 cso [5]
(b)	$7=-8+3d \Rightarrow 7=-8+3\left(\frac{4k+9}{5}\right) \Rightarrow 25=4k+9 \Rightarrow k=4$ *	M1A1 cso [2]
(c)	$\left[d=\frac{4\times4+9}{5}=5 \quad \text{you may see this in part (c)}\right]$ $\frac{n}{2}(2\times-8+(n-1)\times5)=5[-8+(n+10-1)5]+105$ $\Rightarrow -21n+5n^2=370+50n+210 \Rightarrow 5n^2-71n-580=0$ $\Rightarrow (5n+29)(n-20)=0 \Rightarrow n=20$	M1 dM1 M1A1 [4]
		Total 11 marks